Development of Data Analyst-Like Machine Learning Artificial Intelligence Software

Introduction

Today, massive amounts of information are gathered every single moment—from people shopping online and posting on social media, to weather reports and science experiments. To most people, this information looks confusing, like endless pages of numbers and symbols scattered across rows and columns. Without explanation, it feels meaningless. But for someone who studies data, those numbers are like hidden stories. With patience and the right skills, patterns can be discovered, connections can be made, and the data can explain things about how people behave, how nature works, or how businesses grow.

In this situation, a computer program has been designed to act like a human data analyst. It is not just a simple classroom project or a calculator, but a system that tries to copy the complicated thinking process of a real analyst. Its job is to take raw, messy information, clean and organize it, study it carefully, and then share the results in ways people can easily understand. Unlike live data that changes every second, this program can only work with .csv files, a type of file that stores data in a table format. By focusing on .csv files, the program can fully study completed datasets without worrying about information that is always changing.

This is made possible because of Python, a computer language that is considered the “universal language” of data science today. Python is popular not only because its words and commands are easy to read, but also because it has a huge collection of special tools, called libraries. These libraries are like toolboxes created by experts, which allow programmers to do very complex things with just a few lines of code.

Thanks to these libraries, the program does not have to start from scratch by writing thousands of lines of math code. Instead, it can focus on bigger tasks—like figuring out the best method to study a dataset, adjusting to different types of problems, and showing the results in simple and clear ways.

Some libraries are especially important in this process. NumPy provides fast ways to handle large sets of numbers, forming the basic building blocks for calculations. Pandas builds on this by offering the DataFrame, a table-like tool that organizes messy data into a clean format that can be filtered, grouped, and transformed. Together, NumPy and Pandas give the program the ability to prepare raw data for deeper study.

After the data is prepared, the program has to “think.” This is where Scikit-learn, or Sklearn, becomes vital. Sklearn provides many machine learning models—methods that allow computers to learn from examples, spot patterns, and make predictions. Some of these methods are supervised, which means they learn from data that already has answers provided. Others are unsupervised, meaning they look for hidden groups or patterns without being told what to find. By using Sklearn, the program does not need to create every method from the ground up, but instead can choose from ready-made options and focus on deciding which one fits the data best.

For example, if the program is given a .csv file about toy sales, it could use regression models to predict how many toys will be sold in the future, classification models to mark toys as “popular” or “unpopular,” or clustering methods to group toys that are similar. Beyond simply choosing a method, the program can test itself by splitting the data into training and testing sets, changing settings, and checking how accurate its answers are. This careful testing process is much like how a human analyst tries out different ideas before deciding on the best one.

The last step is communication, because numbers alone can be hard for most people to understand. To solve this, the program uses Matplotlib and Seaborn, two Python libraries that specialize in making charts and graphs. These tools turn results into colorful pictures that highlight trends, such as toy sales rising in December or how lowering prices can increase sales. By turning numbers into images, the program makes sure the results are not only correct but also easy for people to follow.

The importance of this project is not only in how it works, but also in what it shows about technology today. With the right tools and creativity, difficult jobs that once required expert teams can now be made easier through accessible software. While challenges will always appear—like incomplete data, mistakes in the code, or limits in experience—each problem gives a chance to learn and improve.

On a bigger scale, this project proves the power of modern technology. Python and its libraries make it possible for anyone determined enough to explore advanced fields like data science and machine learning. What once needed years of training and expensive resources can now be tried with just a computer and persistence. The program demonstrates how technology can copy the thinking of human analysts and make useful knowledge from messy data.

In the end, creating a program that acts like a human analyst. By focusing on .csv files and using Python libraries like NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn, the system can clean data, study it, and present insights automatically. More than just a coding project, it shows how determination and modern tools can turn confusing information into clear, meaningful information.

Purpose and Objectives

The main goal of this study is to make an easy computer tool using Python that works like a helper for studying data. It’s like having a virtual data expert. This tool will help people who don’t know much about data, or who don’t have strong computers, to quickly clean, study, and understand .csv files (which are like tables of information). It will also make pictures and charts so the data is easier to understand.

This study has a few main goals:

1. Build the system – Make a Python program with different parts: one that takes in the file, one that cleans the data, one that studies the data, and one that makes pictures (like charts and graphs) from it.

2. Fix messy data – Add smart methods so the program can handle missing pieces, mistakes, and weird numbers in the data.

3. Make it automatic – Teach the program how to figure out what type of data it's looking at and choose the best way to study it.

4. Show results clearly – Create an easy-to-read display so anyone can understand the charts and graphs, even if they're not data experts.

5. Test and improve – Try the system on different kinds of files, see how well it works, and explain what it can't do, especially with tricky or very messy data.

Statement of the Problem

The big problem in this study is that there isn't one easy tool that everyone can use to clean up, study, and show results from .csv files (those are files with rows and columns, kind of like a spreadsheet). Right now, it's really hard because people need special skills, the tools that work fast use up a lot of computer power, and the data often has mistakes or missing parts. This study wants to make a simple system that can fix those problems, so even beginners with normal computers can use it.

To guide the development and evaluation of this program, the study will address the following research questions:

1. How can a Python-based program be developed to automatically clean, analyze, and visualize data from .csv files?

2. What methods can the program apply to handle missing values, errors, or inconsistencies in the dataset without compromising accuracy?

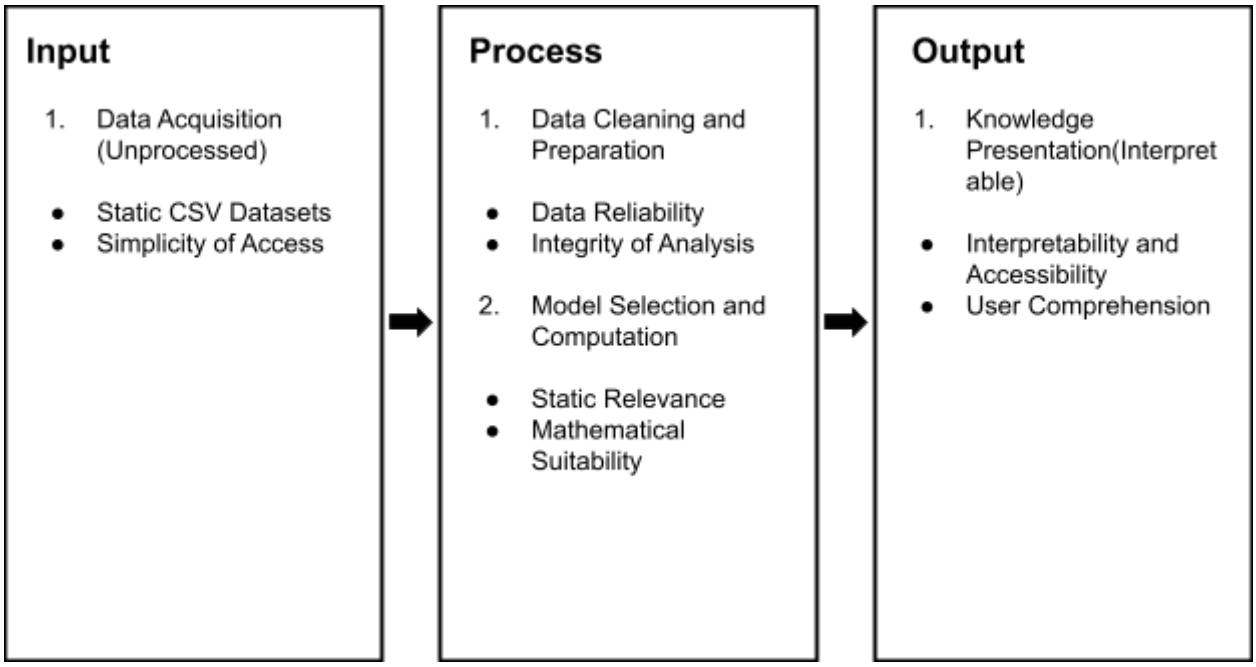
3. How effective is the program in selecting appropriate analytical models for different types of datasets

4. In what ways can the program present results (e.g., through charts and graphs) to make them easily understandable to non-expert users?

5. What are the limitations of the system in handling qualitative datasets and highly inconsistent data, and how do these affect its overall performance?

Conceptual Framework

This study uses a special plan called a framework to show how the system will work. It's like a flow that goes Input → Process → Output, but changed a little to fit how the data tool works. The plan is called the Automated Data Curation and Insight Generation Model (ADCIG). It makes sure the tool isn't just a simple program, but a smart system that follows the best ways to study data. The two main ideas that guide it are: making things automatic and making them easy for everyone to use.



This study shows the system as a step-by-step pipeline, kind of like a factory line that turns messy data into useful knowledge.

1. Input (Step 1): It starts with the raw .csv file, which might be messy or have missing pieces.
2. Process (Step 2): The data goes into the special engine, where it gets cleaned up, studied, and turned into numbers and results.
3. Output (Step 3): The results are shown as easy-to-read charts and summaries that the user can understand.

This way, the system takes a simple file and, using smart methods, gives back clear and helpful information.

Significance of the Study

This study is important because it makes data analysis easier for everyone, not just experts. Right now, most tools are too hard or too very common), this study gives people a tool

that is practical and doesn't need a powerful computer. It's like giving everyone a chance to understand and use data, not only big companies.

The biggest importance of this study is that it makes studying data something that anyone can do, not just experts. Normally, working with data can be really hard, expensive, and requires people to know complicated math or programming. This study removes those barriers by creating a tool that makes data analysis easy, affordable, and useful for everyone.

Empowering Students, Teachers, and Researchers: With this tool, people don't need to learn complicated computer languages like Python or R or master difficult software just to understand data. Students can use it to finish school projects or prepare for science fairs more easily. Teachers can use it in classrooms as a simple and friendly way to show students how to study data. Researchers can use it to quickly get clear insights without wasting time on tricky setup steps. The focus shifts away from "how to process the data" to "what the data is telling us," which saves time and makes the whole process less stressful.

Helping Small Businesses and Non-Profits: These groups often don't have enough money to hire experts or buy expensive programs for analyzing data. That makes it harder for them to keep up with bigger companies that have more resources. This system gives them a free and efficient way to turn their everyday information, like customer feedback, product logs, or donation records, into helpful knowledge. They can spot important patterns, understand what's working well, and check if their plans or programs are successful. This means they get the same kind of smart decision-making power that bigger and richer organizations already have, putting them on a more equal level.

Definition of Terms

Automated Analysis – This means the program can study the data all by itself. It chooses the best method (like finding the average, checking if things are related, or grouping similar items) without the person telling it exactly what to do.

ADCIG Model (Automated Data Curation and Insight Generation Model) – This is the plan the system follows. It's like a step-by-step path: first you put in the raw data (Input), then the program cleans and fixes it (Data Curation), studies it (Automated Analysis), and finally shows it in pictures and graphs (Visualization).
CSV File (Comma-Separated Values) – This is the type of file the program uses. It's a simple text file that looks like a table. Columns are split by commas, and each row is written on a new line.

Data Curation – This is the cleaning step. The program fixes missing pieces, makes everything follow the same format, and finds mistakes in the data so it's ready to be studied.

Data-Driven Decision-Making – This means making choices based on what the numbers and charts say, instead of just guessing or relying on stories.

DataFrame (Pandas) – This is the main tool the program uses to hold the CSV data. Think of it like a big table with rows and columns that you can change and organize.

Libraries (Python) – These are ready-made sets of code that help the program do special jobs quickly. For example: NumPy for math, Pandas for handling tables of data, Scikit-learn for machine learning, Matplotlib and Seaborn for making graphs and charts.

Model Selection – This is when the program decides the best way to study the data. For example, it might use regression to predict something, or clustering to group similar items.

Non-Expert User – This is the kind of person or group the system is made for. They don't know much about advanced math, coding, or fancy software, but they still need to study data.

Scikit-learn (Sklearn) – A Python library that helps the program study data using machine learning. It has tools for things like prediction, classification, and grouping.

Visualization – The last step, where the program takes all the results and shows them as easy-to-understand pictures (charts and graphs) so anyone can get the meaning quickly.